Presentation Prep (+ Follow Up changes)

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November 25th, 2024

Introduction

One of the questions raised in the Michaelmas presentation was regarding the link function and the expectation term in the Generalised Additive Models (GAMs) equation. These were used as an intermediate step to build up to Additive Gaussian Process Models from the linear additive case. The following notes aim to address this question with some clarification. GAMs extend additive models by allowing for nonlinear relationships between predictors and the response variable while maintaining interpretability. The flexibility of GAMs is achieved by incorporating smooth functions of predictors. This report focuses on the role of the link function, the conditional probability distribution of the response variable, and the expectation formulation within GAMs.

General Framework

The general form of a generalised additive model is given by:

$$g\big(E[y \mid \mathbf{X}]\big) = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p),$$

where:

- y: Response variable.
- $\mathbf{X} = (X_1, X_2, \ldots, X_p)$: Predictor variables.
- $E[y \mid \mathbf{X}]$: Conditional expectation of y given $\mathbf{X}$.
- $g(\cdot)$: Link function that relates the conditional mean $E[y \mid \mathbf{X}]$ to the additive predictor.
- β_0: Intercept term (previously referred to as the 'bias')
- $f_1, f_2, \ldots, f_p$: Smooth functions of predictors, capturing nonlinear effects.

The additive structure allows each predictor to contribute independently to the response, simplifying the interpretation and ensuring that the model remains interpretable even when incorporating complex, nonlinear relationships.

The Role of the Link Function

The link function $g(\cdot)$ maps the conditional mean $E[y \mid \mathbf{X}]$ to the scale of the additive predictor:

$$g\big(E[y \mid \mathbf{X}]\big) = \eta,$$

where η is the linear predictor:

$$\eta = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p).$$

Common choices of link functions depend on the assumed distribution of y:

- **Identity Link** ($g(\mu) = \mu$): For normal responses.
- **Log Link** ($g(\mu) = \log(\mu)$): For Poisson responses.
- **Logit Link** ($g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$): For binomial responses.

Selecting the appropriate link function is crucial as it ensures that the model's predictions are valid and interpretable. The link function transforms the expected value of the response variable to the scale where the additive predictors operate, facilitating the modeling of various types of data.

Conditional Probability and Expectation

The response variable y is assumed to follow a distribution from the exponential family, characterised by its conditional probability density or mass function $p(y \mid \mathbf{X})$. The exponential family encompasses a wide range of distributions, including normal, binomial, Poisson, and gamma distributions. This generality allows GAMs to be applied to diverse types of data by appropriately modeling the response variable's distribution.

The conditional expectation of y given $\mathbf{X}$ is expressed as:

$$E[y \mid \mathbf{X}] = \int_{-\infty}^{\infty} y \cdot p(y \mid \mathbf{X})\, dy.$$

For discrete response variables (e.g., counts or binary outcomes), the expectation becomes a summation:

$$E[y \mid \mathbf{X}] = \sum_{y} y \cdot p(y \mid \mathbf{X}).$$

Key Features of GAMs

- **Flexibility**: By using smooth functions $f_i(X_i)$, GAMs capture nonlinear relationships between predictors and the response variable.
- **Interpretability**: Each function $f_i(X_i)$ represents the isolated effect of predictor X_i, holding other predictors constant.
- **Distributional Assumptions**: GAMs extend by modeling responses from the exponential family (e.g., normal, Poisson, binomial).
- **Prediction Constraints**: The link function ensures that the predicted mean $E[y \mid \mathbf{X}]$ remains within the valid range of the response variable (e.g., probabilities are constrained between 0 and 1).
- **Smoothness Penalty**: GAMs incorporate penalties on the smooth functions to control their flexibility, thereby preventing overfitting and ensuring generalisable models.

Why the Link Function is Applied to the Expectation of the Response Variable

In Generalised Additive Models (GAMs) and Generalised Linear Models (GLMs), the link function $g(\cdot)$ plays a crucial role in connecting the predictors to the response variable. A fundamental design choice in these models is to apply the link function to the **conditional expectation** $E[y \mid \mathbf{X}]$ rather than to the response variable y itself. I believe this was the key point of clarification that was being questioned in the presentation itself and this sections aims to address that.

Modeling the Central Tendency

Expectation vs. Individual Observations:

- **Stability and Interpretability**: $E[y \mid \mathbf{X}]$ represents the central tendency or mean of the response variable given the predictors. Modeling the mean provides a stable and interpretable summary of the relationship between predictors and the response, especially in the presence of variability or noise in the data.

- **Simplification**: Focusing on the expectation reduces the complexity of the model. Directly modeling individual observations y would necessitate accounting for the entire distribution of y, including its variance and higher moments, complicating both the model structure and the estimation process.

Flexibility Across Diverse Distributions

Non-Normal Distributions: Response variables in real-world applications often do not follow a normal distribution. For instance, binary outcomes (e.g., success/failure) or count data (e.g., number of events) require different modeling approaches. Applying the link function to $E[y \mid \mathbf{X}]$ allows GAMs to flexibly handle a wide range of distributions from the exponential family, such as binomial, Poisson, and gamma distributions. Applying the link function directly to y would make it challenging, if not impossible, to enforce such constraints on individual observations, leading to invalid or nonsensical predictions.

Illustrative Example

Consider a binary response variable y representing the success or failure of a treatment. Suppose we aim to model the probability of success based on a predictor X.

Without a Link Function: Attempting to model y directly would involve predicting binary outcomes, which does not naturally facilitate the modeling of probabilities and would lack a mechanism to enforce that predicted probabilities lie within the interval $[0, 1]$.

With a Logit Link: Applying the logit link to the expectation yields:

$$\log\left(\frac{E[y \mid X]}{1 - E[y \mid X]}\right) = \beta_0 + f(X)$$

Here, $E[y \mid X]$ represents the probability of success given X. The logit transformation ensures that the linear predictor $\eta = \beta_0 + f(X)$ can take any real value, while $E[y \mid X]$ remains bounded between 0 and 1. This setup allows for a meaningful and interpretable relationship between the predictor and the probability of success.

Summary

Applying the link function to the conditional expectation $E[y \mid \mathbf{X}]$ rather than to the response variable y itself offers several advantages:

- **Stability and Interpretability**: Focuses on modeling the central tendency, providing a clear and stable relationship between predictors and the response.

- **Flexibility**: Accommodates a wide range of distributions and ensures that predictions remain within valid bounds.

- **Linearity on the Link Scale**: Establishes a linear relationship between the transformed mean and the predictors, simplifying estimation and interpretation.

- **Enhanced Estimation and Inference**: Facilitates the use of Maximum Likelihood Estimation and maintains desirable statistical properties of the estimators.

This design choice underpins the effectiveness of GAMs and GLMs in modeling diverse types of data, balancing flexibility with interpretability and statistical rigor.

Conclusion

Generalised additive models provide a powerful framework for flexible and interpretable statistical modeling. The link function plays a central role in ensuring valid predictions and enabling the use of various response distributions. By combining the link function with smooth functions of predictors, GAMs allow for the discovery of complex relationships between predictors and the response variable.

References

- J. Hastie and R.J. Tibshirani. Generalized additive models. Chapman & Hall/CRC, 1990.
- M. Clark. General Additive Models